

Department of Electrical & Electronic Engineering
Brac University
Semester: Spring 25



EEE101L

ELECTRICAL CIRCUITS I LABORATORY

Section:05

Experiment Name: Verification of Thevenin's Theorem and Maximum Power Transfer Theorem.

Prepared by: Souvik Barman Ratul.

Group Number: 05

Group members:

SL	Student ID	Name
01.	24104215	Shreya Das
02.	24121205	Souvik Barman Ratul
03.	24121280	Tanisha Hannan
04.	24121308	Alif Tamjid

Part-A: Verification of Thevenin's Theorem

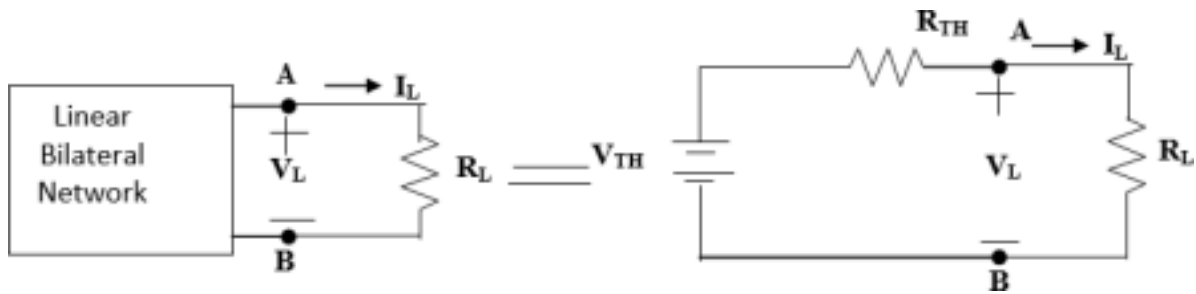
1. **Objective**: To verify Thevenin's theorem with reference to a given circuit.

2. **Theoretical Background**: It is often desirable in circuit analysis to study the effect of changing a particular branch element while all other branches and all the sources in the circuit remain unchanged. Thevenin's theorem is a technique to this end and it reduces greatly the amount of computations which we have to do each time a change is made. Using Thevenin's theorem, the given circuit, excepting the particular branch to be studied, is reduced to the simplest equivalent circuit possible and then the branch to be changed is connected across the equivalent circuit.

The Thevenin's theorem states that any two terminal linear bilateral network containing sources and passive elements can be replaced by an equivalent circuit consisting of a voltage source V_{th} in series with resistor R_{th} where

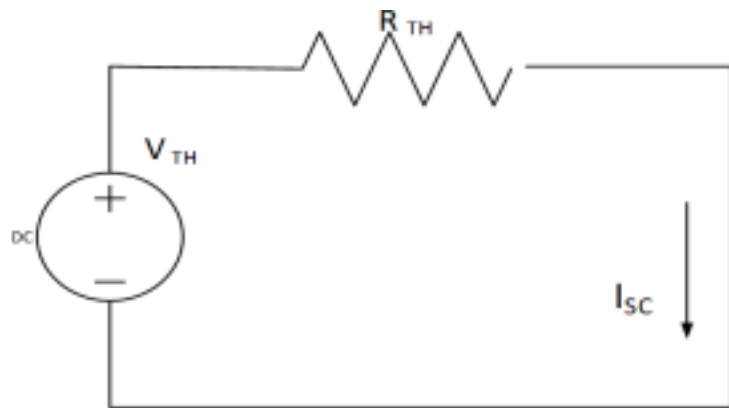
V_{th} = the open circuit voltage (VOC) at the two terminals A & B.

R_{th} = the resistance looking into the terminals A and B of the network with all sources removed.



There are several methods for determining Thevenin resistance R_{TH} . An attractive method for determining R_{TH} is: (1) determine the open circuit voltage, and (2) determine the short circuit current I_{SC} as shown in the figure; then

$$R_{TH} = \frac{V_{OC}}{I_{SC}}$$



3. Equipment:

- Resistors
- Multimeter
- DC Power Supply adjusted to 20V DC
- Breadboard

Circuit Diagram:

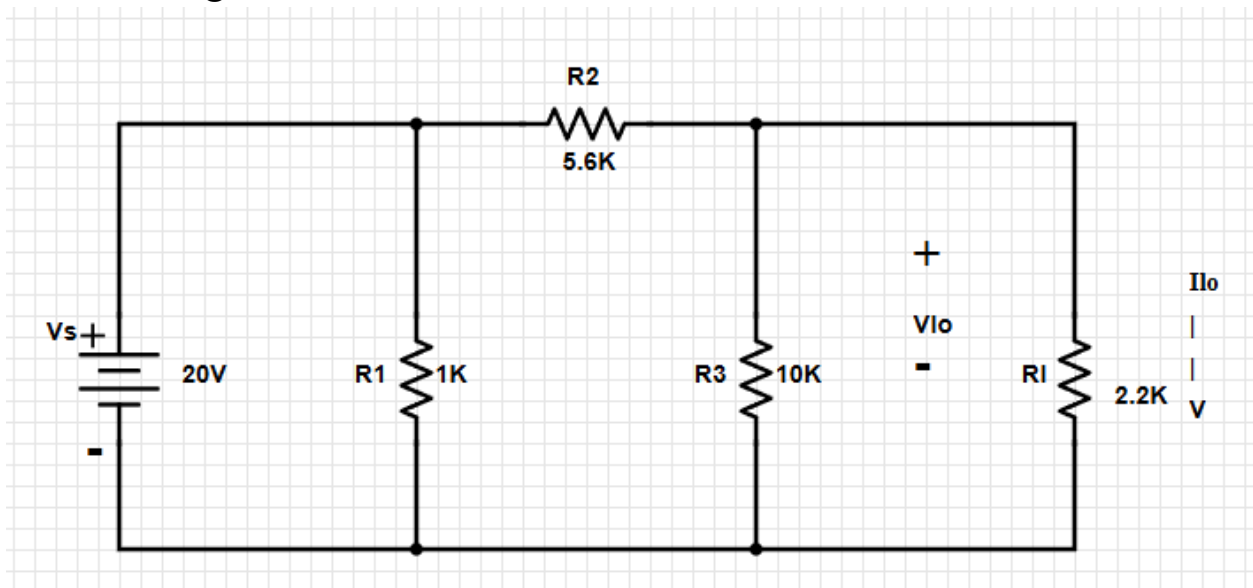


Fig. 1: Original Circuit

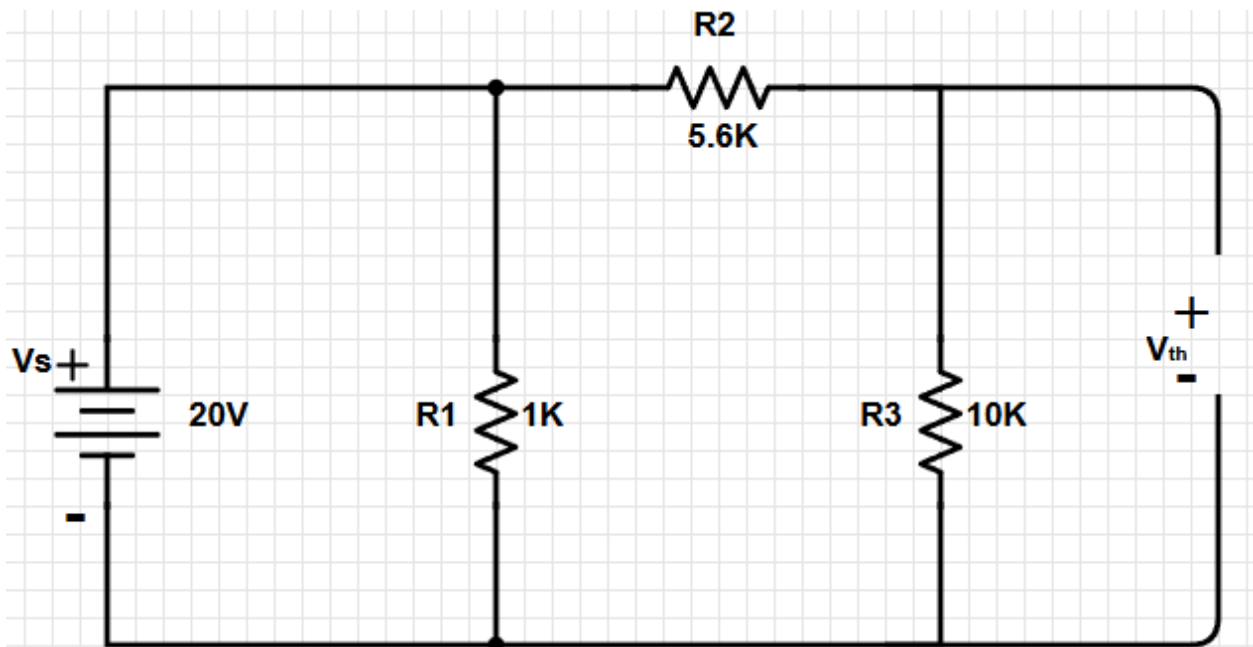


Fig. 2: Circuit For finding V_{th}

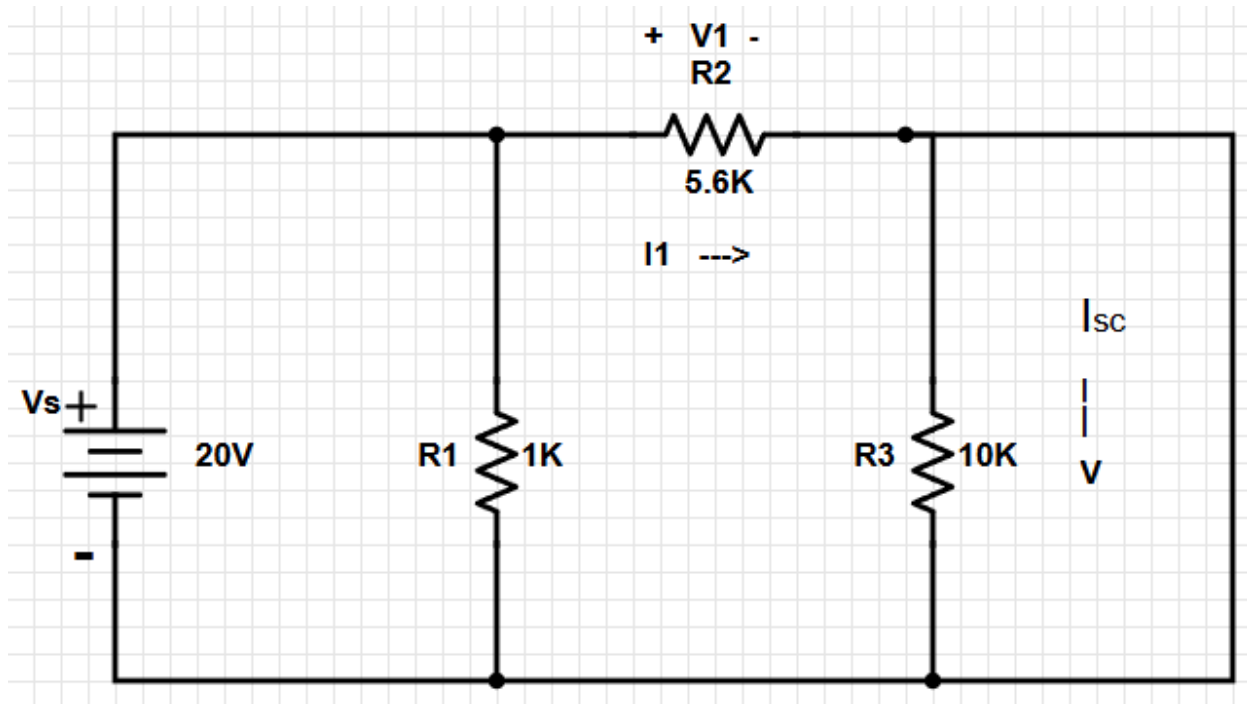


Fig. 3: Circuit for finding I_{sc}

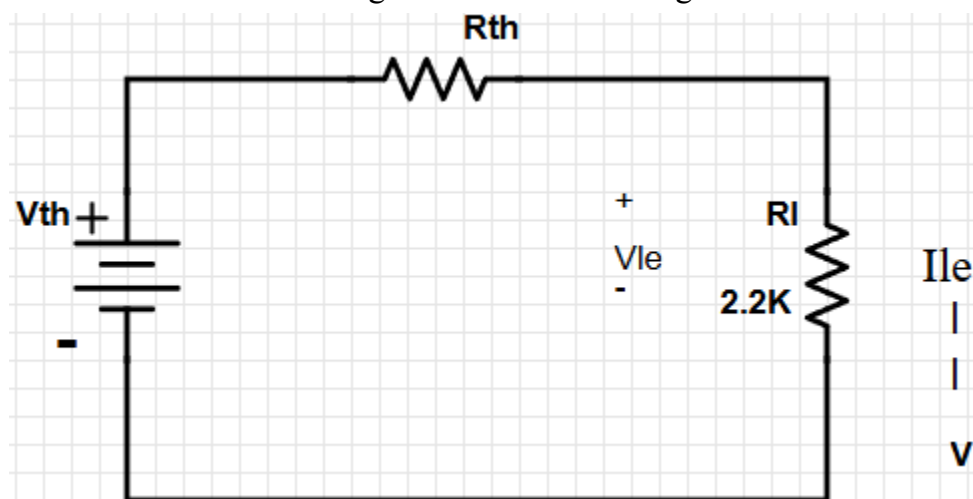


Fig. 4: Equivalent circuit

Data Table:-

Table 1:- Data for Original Circuit

$R_1(K\Omega)$	$V_{I0}(V)$	$I_{I0}(mA)$
2.1	4.8	2.285

Table 2:- Data for V_{th}

$V_{th}(V)$
12.82

Table 3:- Data for R_{th}

Indirect Method:

$R_2(K\Omega)$	$V_1(V)$	$I_1=V_1/R_2(mA)$	$I_{sc}=I_1(mA)$	$R_{th}=V_{th}/I_{sc}(K\Omega)$
5.5	19.8	3.6	3.6	3.561

Direct Method:

$R_{th}(K\Omega)$
3.5

$$\text{Error} = \{(3.561 - 3.5)/3.5\} \times 100\% = (0.061/3.5) \times 100\% = 1.74\%$$

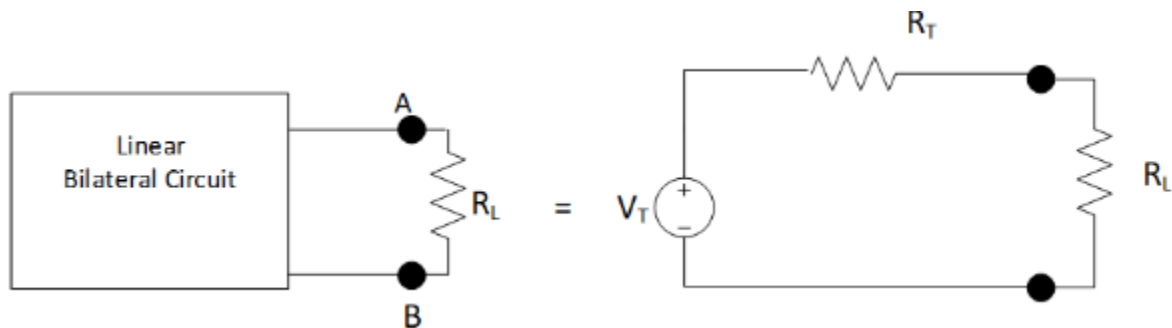
Discussion:-

The experiment successfully verified Thevenin's Theorem, with a minimal 1.74% error due to instrument mistake, resistor tolerances, power supply fluctuations, and minor human errors. These factors caused slight deviations in measured values, but the results remained close to theoretical expectations, confirming that complex circuits can be effectively simplified using Thevenin's equivalent model.

Part-B: Maximum Power Transfer Theorem

1. **Objective:** The objective of this experiment is to verify maximum power transfer theorem.

2. **Theoretical Background:** The maximum power transfer theorem states that a resistive load will receive maximum power when its total resistive value is exactly equal to the Thevenin's resistance of the network as seen by the load.



We know that any linear circuit terminated with a load R_L can be reduced to its Thevenin's equivalent. Now according to this theorem, the load R_L will receive maximum power when

$$R_L = R_{th}$$

3. Equipment:

- Digital Multimeter
- DC power supply adjusted to 20V DC
- Resistor and potentiometer
- Wires

Circuit Diagram:

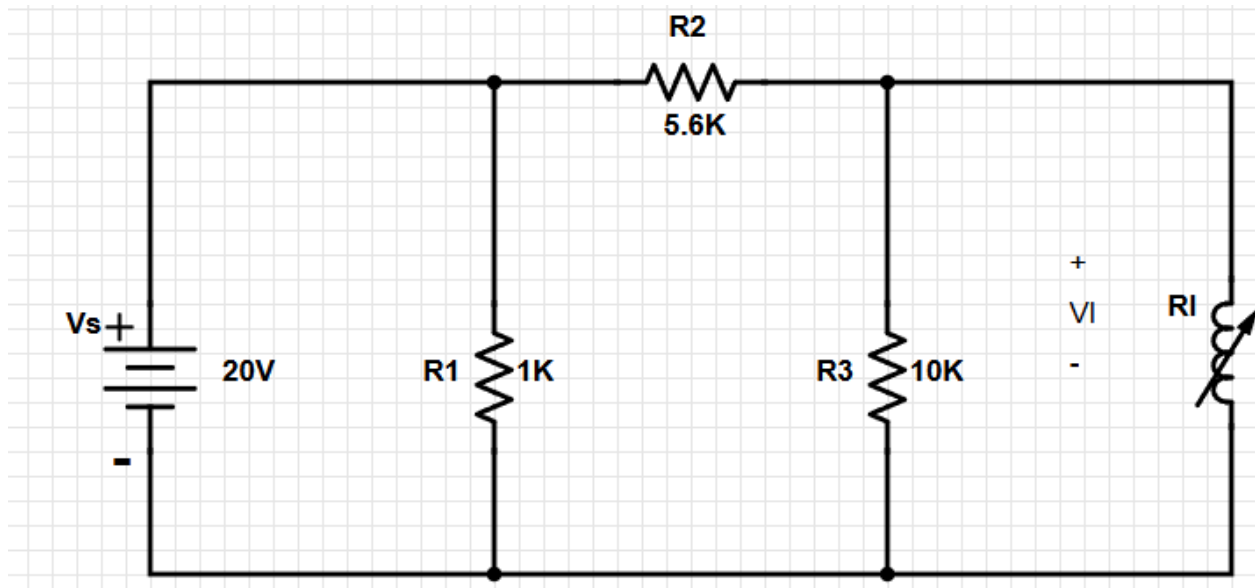


Fig. 5: Circuit for verifying maximum power transfer theorem

Data Table:-

$R_L(K\Omega)$	$V_1(V)$	$P_1 = (V_1^2)/R_L \text{ (mW)}$
1	2.8	7.84
2.2	4.9	10.913
3.2	6.1	11.628
3.5	6.6	12.445
4.6	7.2	11.269
5.5	7.8	11.061
10	9.4	8.836

According to the Maximum Power Transfer Theorem, maximum power should be transferred when **$R_L = R_{th}$** .

From Part A, the Thevenin resistance was found to be **$R_{th} = 3.5K\Omega$**

From the data table, the power is:

At **$R_L = 3.5K\Omega$** $\rightarrow P = 12.445mW \rightarrow P = 12.445mW$ (Maximum)

Since the measured peak power aligns closely with theoretical expectations, the error is minimal.

Appendix:

Electrical Circuits I Laboratory EEE 101L Department of Electrical & Electronic Engineering (EEE) Brac University

Experiment No. 5

Verification of Thevenin's Theorem and Maximum Power Transfer Theorem

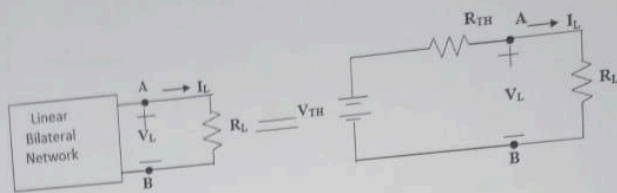
Part-A: Verification of Thevenin's Theorem

1. **Objective:** To verify Thevenin's theorem with reference to a given circuit.
2. **Theoretical Background:** It is often desirable in circuit analysis to study the effect of changing a particular branch element while all other branches and all the sources in the circuit remain unchanged. Thevenin's theorem is a technique to this end and it reduces greatly the amount of computations which we have to do each time a change is made. Using Thevenin's theorem, the given circuit, excepting the particular branch to be studied, is reduced to the simplest equivalent circuit possible and then the branch to be changed is connected across the equivalent circuit.

The Thevenin's theorem states that any two terminal linear bilateral network containing sources and passive elements can be replaced by an equivalent circuit consisting of a voltage source V_{th} in series with resistor R_{th} where

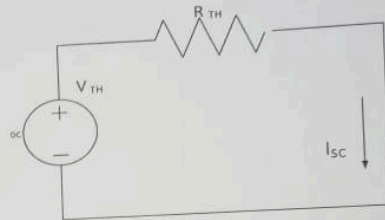
V_{th} = the open circuit voltage (V_{oc}) at the two terminals A & B.

R_{th} = the resistance looking into the terminals A and B of the network with all sources removed.



There are several methods for determining Thevenin resistance R_{TH} . An attractive method for determining R_{TH} is: (1) determine the open circuit voltage, and (2) determine the short circuit current I_{SC} as shown in the figure; then

$$R_{TH} = \frac{V_{OC}}{I_{SC}}$$



3. Equipment:

- Resistors
- Multimeter
- DC Power Supply adjusted to 20V DC
- Breadboard

4. Circuit Diagram:

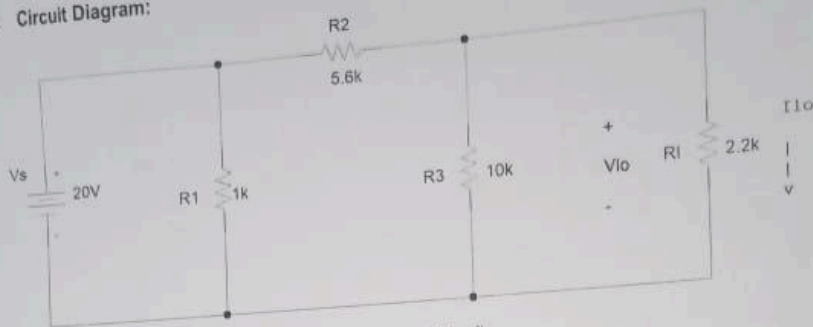


Fig. 1: Original Circuit

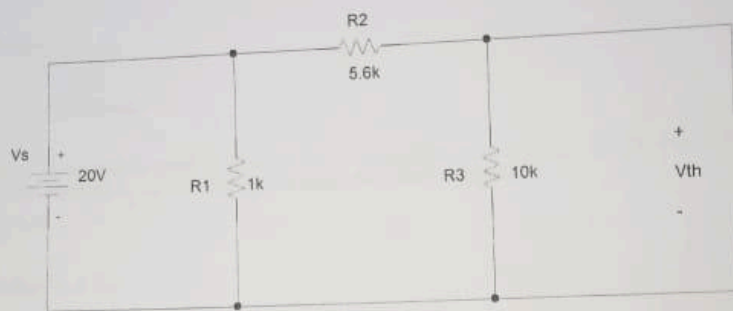


Fig. 2: Circuit for finding V_{th}

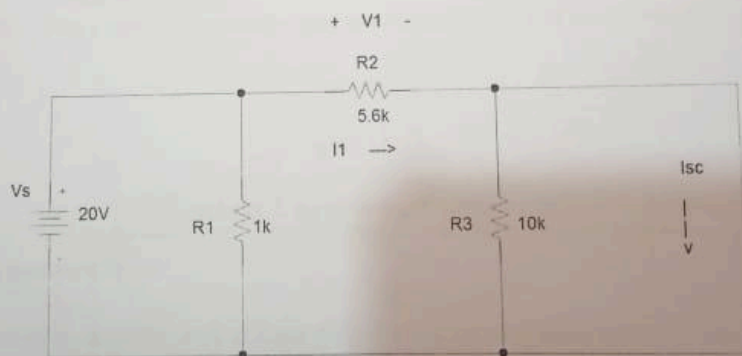


Fig. 3: Circuit for finding I_{sc}

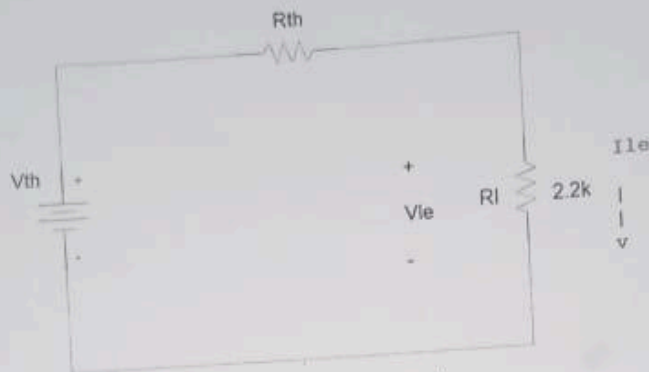


Fig. 4: Equivalent circuit

5. Procedure:

For Original Circuit:

Construct the original circuit as shown in Fig. 1. Connect source voltage $V_s = 20V$ DC from a DC power supply. Measure R_l , V_{l0} , calculate I_{l0} and record the data in the table.

$$I_{l0} = V_{l0}/R_l$$

Finding V_{th} & R_{th} :

- Remove the load resistance R_l and find the open circuit voltage. This voltage is Thevenin voltage i.e. V_{th} .
- Place a short circuit between the open terminals and find the short circuit current I_{sc} .

$$R_{th} = V_{th} / I_{sc}$$

$$I_{sc} = I_1 = V_1 / R_2$$

Get back to the circuit in Fig. 2. Deactivate the DC voltage source. Measure the resistance between the open circuit terminals via the multimeter. Verify the earlier value of R_{th} .

For Equivalent Circuit:

Construct the equivalent circuit as shown in Fig. 4 setting the power supply at V_{th} volts and resistor R_{th} as close as possible to the value found from above. Now measure V_{le} and calculate I_{le} . Compare these values with previous values.

6. Data Table:

Table 1: Data for original circuit

R_1 (k Ω)	V_{I_0} (V)	I_{I_0} (mA)
2.3	4.6	2.265

Table 2: Data for V_{th} *CASE 1:*

V_{th} (V)
12.92

$R_{th} = 3.5$

Table 3: Data for R_{th}

Indirect Method:

R_2 (k Ω)	V_1 (V)	$I_1 = V_1 / R_2$ (mA)	$I_{sc} = I_1$ (mA)	$R_{th} = V_{th} / I_{sc}$ (k Ω)
5.5	19.8	3.6	3.6	3.561

Direct Method:

CASE 2

R_{th} (k Ω)
3.5

Table 4: Data for equivalent circuit

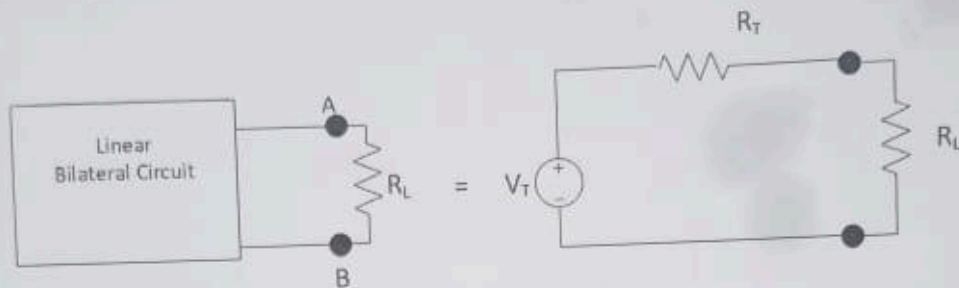
V_{I_e} (V)	I_{I_e} (mA)

Rad 06/07/20
Faculty Signature and Date

Part-B: Maximum Power Transfer Theorem

1. **Objective:** The objective of this experiment is to verify maximum power transfer theorem.

2. **Theoretical Background:** The maximum power transfer theorem states that a resistive load will receive maximum power when its total resistive value is exactly equal to the Thevenin's resistance of the network as seen by the load.



We know that any linear circuit terminated with a load R_L can be reduced to its Thevenin's equivalent. Now according to this theorem, the load R_L will receive maximum power when

$$R_L = R_{Th}$$

3. Equipment:

- Digital Multimeter
- DC power supply adjusted to 20V DC
- Resistor and potentiometer
- Wires

4. Circuit Diagram:

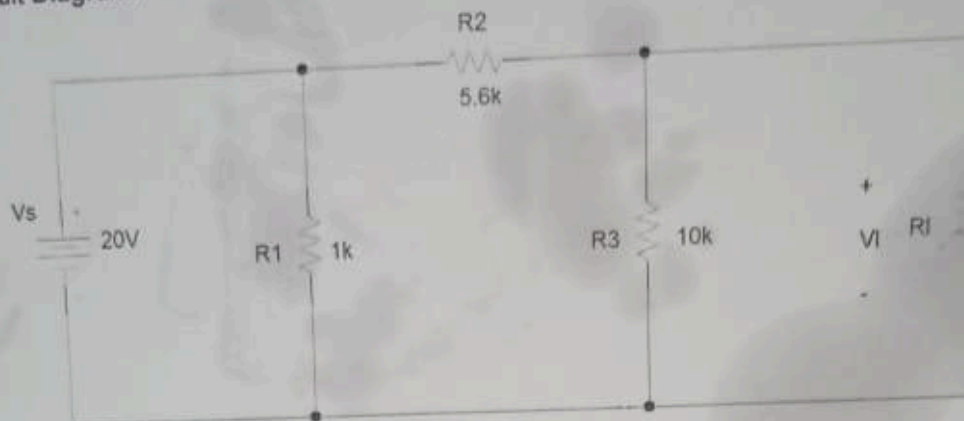


Fig. 5: Circuit for verifying maximum power transfer theorem

5. Procedure:

- Set up the circuit as shown in Fig. 5.
- Use the following values of R_L .

$1k\Omega$, $2.2k\Omega$, $3.3k\Omega$, R_{th} , $4.7k\Omega$, $5.6k\Omega$ and $10k\Omega$

Set R_{th} up using potentiometer. Measure R_L , V_L and calculate $P_L = V_L^2 / R_L$. The power shall be maximum for condition where the $R_L = R_{th}$.

6. Data Table:

R_L ($k\Omega$)	V_L (V)	$P_L = V_L^2 / R_L$ (mW)
1	2.8	7.84
2.2	4.9	10.915
3.3	6.1	11.628
4.7	6.6	12.443
5.6	7.2	11.267
10	9.21	2.856

Faculty Signature and Date

